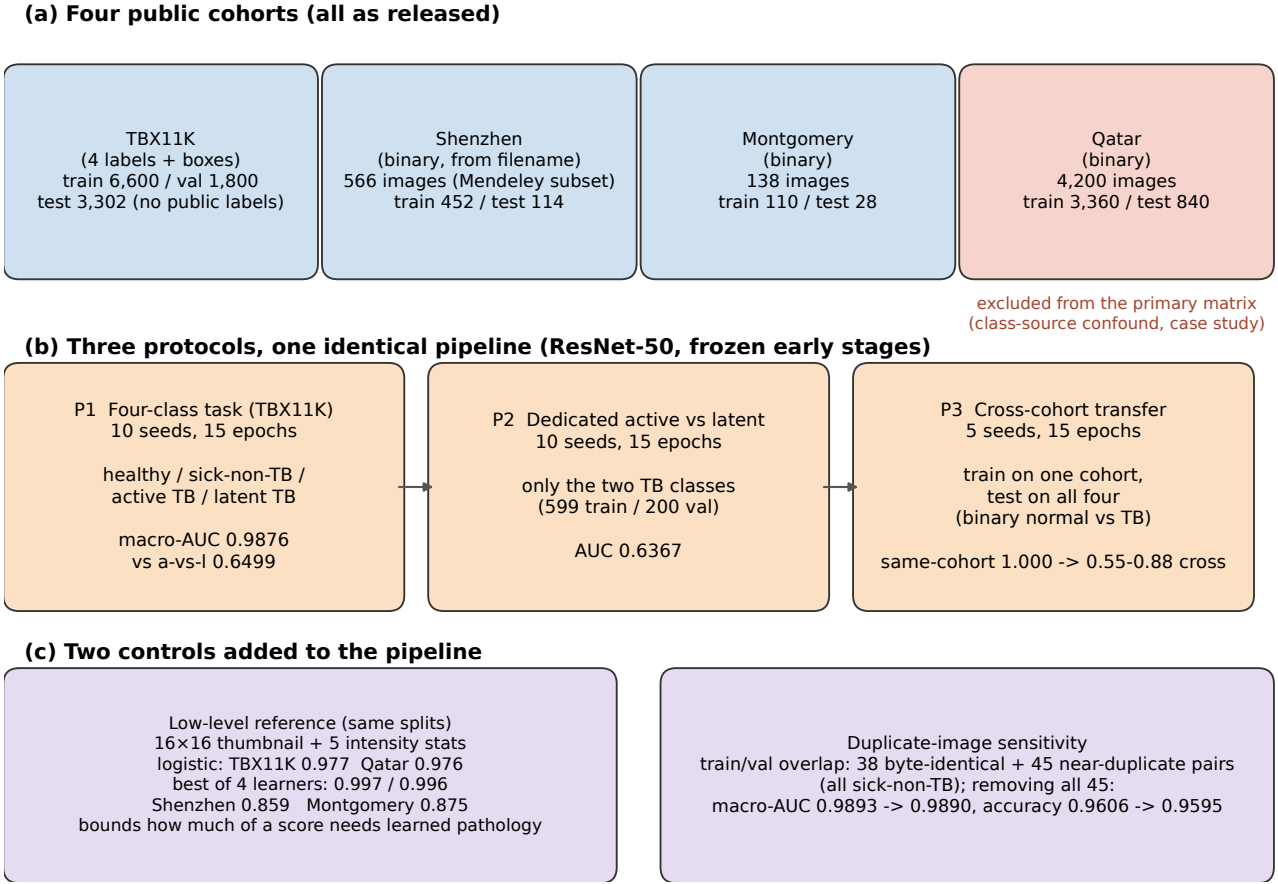
(a) Four public cohorts (all as released)
TBX11K
(4 labels + boxes)
train 6,600 / val 1,800
test 3,302 (no public labels)
Shenzhen
(binary, from filename)
566 images (Mendeley subset)
train 452 / test 114
Montgomery
(binary)
138 images
train 110 / test 28
Qatar
(binary)
4,200 images
train 3,360 / test 840
excluded from the primary matrix
(class-source confound, case study)
(b) Three protocols, one identical pipeline (ResNet-50, frozen early stages)
P1 Four-class task (TBX11K)
10 seeds, 15 epochs
healthy / sick-non-TB /
active TB / latent TB
macro-AUC 0.9876
vs a-vs-l 0.6499
P2 Dedicated active vs latent
10 seeds, 15 epochs
only the two TB classes
(599 train / 200 val)
AUC 0.6367
P3 Cross-cohort transfer
5 seeds, 15 epochs
train on one cohort,
test on all four
(binary normal vs TB)
same-cohort 1.000 -> 0.55-0.88 cross
(c) Two controls added to the pipeline
Low-level reference (same splits)
16×16 thumbnail + 5 intensity stats
logistic: TBX11K 0.977 Qatar 0.976
best of 4 learners: 0.997 / 0.996
Shenzhen 0.859 Montgomery 0.875
bounds how much of a score needs learned pathology
Duplicate-image sensitivity
train/val overlap: 38 byte-identical + 45 near-duplicate pairs
(all sick-non-TB); removing all 45:
macro-AUC 0.9893 -> 0.9890, accuracy 0.9606 -> 0.9595